Industrial Electronics N5

Module 4
Integrated circuits and transducers

Learning Outcomes

At the end of this module, learners will be able to:

Section 4.1 integrated circuits

- Explain in words what an integrated circuit is
- List and explain the operating principles of IC packages
- Explain the term integrated bridge rectifiers
- List the different types of bridge rectifier packages
- List advantages and disadvantages of integrated bridge rectifiers
- Explain with the aid of a neat labelled circuit diagram what is an integrated voltage regulator
- Explain in words the term electrostatic discharge
- Explain the damages that can be caused by electrostatic discharge
- List and explain modes in which ESD damage occurs
- Explain how you should handle circuits boards
- Explain what CMOS device is and show how CMOS device protection is achieved

Section 4.2 transducers

- Explain in words the term transducer
- List the points to be considered when selecting a transducer
- List the requirements a transducer must meet to ensure reliable operation of a control system.
- Draw a neatly block diagram of how transducers can be classified.
- List the requirements a transducer must meet to ensure reliable operation of a control system.
- Draw a neatly block diagram of how transducers can be classified.
- List and explain the functions of TWO types of transducers.
- List the types of passive transducers.
- Explain in words what is meant by the term potentiometer.
- Explain the operating principle of a potentiometer.
- List, explain the operating principles or circuit diagrams applications and draw a neatly labelled circuit diagrams / construction of the potentiometric displacement transducers.
- Explain in words what is meant by the term potentiometer resolution.
- Calculate the following values of potentiometer resolution:
    - Percentage resolution
    - Voltage difference of two adjacent turns.
    - Total voltage
    - Total number of turns
    - Stress
    - Strain
- List the different areas of application for a potentiometer resolution.

- Explain in words the following terms:
    - Stress
    - Strain
- Explain in words the term strain gauge.
- Draw a neatly labelled construction of a strain gauge.
- Calculate the following values of a strain gauge
    - Original resistance of the strain gauge
    - Elongation or construction-initiated resistance change
    - Length of the resistance wire
    - Change in length of the resistance wire.
    - Proportional constant
    - Strain
- Draw a neatly labelled construction of a strain gauge.
- Explain the operating principle of strain gauge.
- Explain in words the term gauge factor.
- List the factors that affect the gauge.
- Explain the operating principle and draw a neatly labelled circuit diagram of a strain gauge used in a Wheatstone bridge.
- Explain in words the term thermistor.
- Explain operating principle and draw a neatly labelled construction of a thermistor.
- Calculate the following values given by thermistor
    - A and B are constant
    - Temperature in degree Kelvin
- List, explain the operating procedure, functions, applications and draw neatly labelled circuit / block diagrams / construction of the following transducers:
    - Inductive transducers
    - Voltage and current transducers
    - Photo-electric transducers

4.1 INTEGRATED CIRCUITS

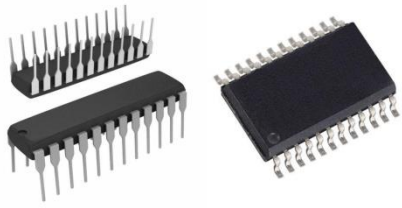
4.1.1 integrated circuits and their applications

What is an integrated circuit?

These are electronic devices made from semiconductor material containing a large number of circuits in a single chip with a number of components such as transistors, resistors, capacitors, and diodes. In short, it is a composition of circuits within a circuit, it is also in short abbreviated as IC. The components inside this chip can range from hundreds to millions or even greater depending on the package.

Integrated circuits do many functions from simple logic functions to complex processing. They are almost in every electronic device such as oscillators, computers, games smart watches and more. Below is a typical example of how integrated circuit looks like;

https://www.allelcoelec.com/blog/dual-inline-package%28dip%29-an-overview.html?srsltid=AfmBOorWslMe2lEj9aE5sP9DEzJ7Fpm3ksMFarxNbtVpOkafypRqGQMq



4.1.2 integrated circuit packages

IC packages serve as support structures and protective shields for the silicon chips that power these electronic devices. They **safeguard** these chips against environmental hazards and regulate heat and facilitate connections with other components. (AnyPCBa, 2025)

Types of IC packages and their applications

- **Dual inline package**: widely used in old computers, video game consoles
- **Surface mounted device** (SMD) : used in consumer electronics such as smart phones and laptops.
- **Ball grid array** (BGA) : used in high performance electronics such as CPU, game consoles because of their ability to handle many connections.
- **Chip on board** (BOG) : mostly found in LED lighting and cheap electronic gadgets where space and cost savings are priorities.
- **Quad flat no-leads** (QFN) : ideal for portable electronics where space is limited and heat management is crucial like in cellphones and media players.

4.1.3 FUCTIONS AND APPLICATIONS OF IC'S

- **Small scale integration** (SSI) – this is where ten to hundreds of transistors are imbedded per chip.
- **Medium scale integration** (MSI) – hundreds to hundred thousand transistors per chip
- **Very large-scale integration** (VLST) – up to million transistors per chip
- **Ultra-large scale** (ULSI) – up billions of transistors per chip.

4.1.4 INTEGRATED BRIDGE RECTIFIER

As studied in unit two, bridge rectifiers are types of rectifiers which use four diodes connected in a bridge to rectifier the input AC voltage to a DC pulsating voltage. In the case of integrated circuits, this bridge circuit is now packaged in an IC package so that it becomes a single device instead of four diodes. These devices are compact, self-contained units to convert AC to DC.



What is the reason for integrated bridge rectifiers?

They are designed to:

- **Save space** since the circuit is reduced to a single unit where the four diodes are imbedded into a single device.
- **Ease of installation**, since these devices come with four pins, the design iis very simple.
- **Provide efficient conversion**, they effectively convert AC to DC with minimum loss of power.

4.1.5 BRIDGE RECTIFIER PACKAGES

- **Schottky bridge rectifiers**: have low forward voltage drop and fast switching capabilities, mostly used in high frequency applications.
- **Surface mounted bridge rectifiers**: they are very compact and designed for surface mounting on PCB's, used in space- constrained applications.
- **Three phase bridge rectifiers**: these are used for high power applications and convert three phase AC input into DC output, they are used in industrial and high power systems.
- **Single phase bridge rectifiers**: most common type designed for low to medium power applications, used for converting single phase AC voltage to DC voltage.

4.1.6 ADVANTAGES AND DISADVANTAGES OF INTEGRATED BRIDGE RECTIFIERS

Advantages; (THOBEJANE, 2024)

- They are **usually small in size** saving space on circuit board
- They have low power requirement since they use diodes
- **Improved reliability** with lower failure rate
- **Reduced cost**, buying a single component cost less than purchasing many discrete components
- It has **good thermal management ability** since some IC packages come with built-in heatsink,

Disadvantages ;

- **Customization is limited** since IC bridge rectifiers come with fixed specifications.
- The **replacement is difficult** in that if one part fails, the whole unit needs to be replaced.
- **Limited availability** and variety since they are designed for specific tasks
- They tend to **develop heat concentration** even though they come with heatsinks

4.1.7 INTEGRATED CIRCUIT VOLTAGE REGULATOR

An integrated circuit voltage regulator is a three terminal semiconductor device used in power supplies to maintain the output DC voltage connected to a load constant regardless of load variations. Below shows the two types of regulator series; the 7900 series and the 7800 series where the 7900 series is a negative output regulator and the 7800 is a positive output regulator. (PROTO SUPPLIES, n.d.)



The 7906 IC regulator



The 7812 IC regulator

4.1.8 ADVANTAGES AND DISADVANTAGES OF IC VOLTAGE REGULATORS

Advantage;

- They are compact in size minimizing space take on PCB
- They are reliable and minimize risk of failure
- They are efficient devices
- They have a built-in protection circuit which offer protection for over voltages and over currents.

Disadvantages;

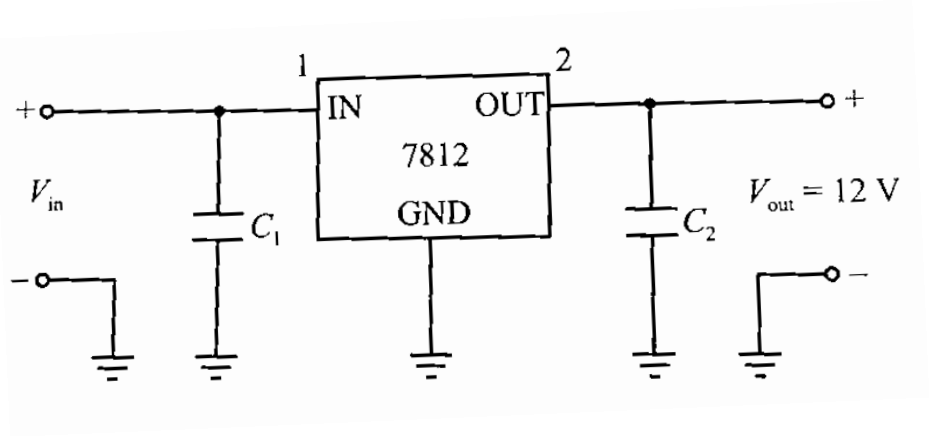
- Linear regulators tend to dissipate heat under heavy loads
- Switching regulators tends to introduce significant noise
- For some applications, IC regulators are expensive
- They offer limited flexibility.

4.1.9 FUNCTIONS AND DIFFERENT TYPES OF IC VOLTAGE REGULATORS (KRAFT, 2024)

- **Fixed positive and negative voltage regulators**: these type deliver a fixed specific output voltage to the load ensuring a stable output to the load.
- **Adjustable voltage regulator**: these offer flexibility, the regulated voltage can now be adjusted within specific limits.
- **Dual-tracking voltage regulator**: it will maintain equal voltage levels both positive and negative, mostly dual power supplies use this type of IC regulator.

4.1.10 THE 7812 VOLTAGE REGULATOR

The **7800 series voltage regulator** is a fixed positive voltage regulator. The first two numbers in 7800 (78) define the type of voltage regulator and the last two numbers (12) define the fixed positive output voltage ie +12V dc. These numbers can go from 7805(+5V) all the way to 7824 (+24V). **The operating principle is as follows**; consider the diagram below

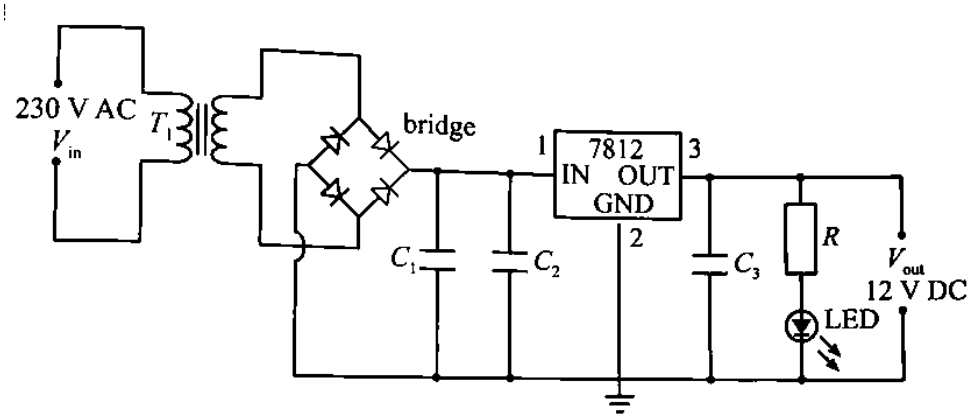


- Un-regulated voltage is linked to the IN(input) terminal of the IC regulator
- The output after regulation will +12V
- **Capacitors** are sometimes used at the input and output side to function as line filter to enhance transient response at the output and prevent undesired oscillation at the input sides.

Example 01 application of 7812 in a power supply

Show how a +12V can be produced using a voltage supply connected to a full-wave bridge rectifier showing all four diodes, a capacitor and the IC regulator. (THOBEJANE, 2024)

Solution



4.1.11 ELECTROSTATIC DISCHARGE(ESD)

Integrated circuits are delicate devices that should always be handled with caution since they are sensitive to a charge know as electrostatic discharge. **Should the IC device be exposed to intense levels of this form of discharge, they may need to be replaced or will function for a short duration then stop working depending on the level of exposure**. This is why this section is important since we will learn about this charge, how it is caused, how it affects IC devices and how to handle IC devices in order to prevent this charge from damaging IC devices.

What is electrostatic discharge?

This is a sudden flow of electricity between two objects that have different charges. This happens when two electrically charged objects like a human body and electronic device

encounter one another, the static electricity between them will discharge rapidly. ESD generated from the human body can be several thousand volts. ESD can damage or destroy IC devices, erase its data or set off explosions or fires. This is because all materials are sources of ESD.

How can ESD damage electronic circuitry?

As current passes through the electronic device, it continuously searches for a path with least resistance to ground in order to equalize potentials. The problem with current is that it will travel on every available path, such through the junctions of a diode on integrated circuits to reach the ground. This current at high values will burn the holes and this will be visible as burn markings on the board, this will degrade the operation of the circuit.

4.1.12 ESD sensitivity

This refers to a measure of the degree in which an electronic device such as IC can be damaged by ESD. This sensitivity can be measured by using an ESD meter test and is expressed in the form of ESD models which assess the survivability of electronic devices to ESD. ESD damage can be defined as catastrophic or latent. Catastrophic failure is whereby the device stops working immediately and latent is whereby the device or circuitry starts to malfunction after some time being exposed to ESD. (KRAFT, 2024)

4.1.13 Handling circuit boards

These are tips that can be followed when working with circuit boards having IC devices to prevent ESD damage:

- Use static floor mates where possible
- Make sure proper grounding system for equipment has low impedance for ESD currents to easily dissipate to the ground.
- Use non-static forming when cleaning PC boards
- Keep all synthetic materials at least 15 cm apart from electronic equipment.

4.1.14 Precautionary measures taken when working with CMOS devices

- Wrist traps should be properly maintained and audited on regular basis to prevent malfunctions
- Do not exceed maximum ratings specified by the data sheet
- All CMOS devices should be transported in materials that are antistatic or conductive
- Nylon or other static generating materials should be kept away from CMOS devices

4.1.15 C-MOS INPUT PROTECTION (THOBEJANE, 2024)

The system designer must always be attentive that C-MOS integrated circuits can be damaged internally if subjected to high electrical fields in the gate oxide regions. Due to the high reactive input (Typically 10^{12} Ohm) the circuit is acceptable for static charges. The oxide layer on the input forms a capacitor with the substrate, which can be charged statically. According to tests, the static voltage a human body invokes when a person walks across a polished floor is between 4 and 15Kv. These charges are destructive as soon as they discharge on the inputs of C-MOS circuits because only 100V is required to cause permanent damage. (Deventoer, 2001)

Some integrated circuits use diodes on the inputs to protect circuits. The diodes are connected in a reverse biased configuration between the input (gate) and the positive supply. In other cases, two Zener-diodes are connected as indicated in FIGURE B Voltage pulses which exceed +/-15V are chopped. (Deventoer, 2001)

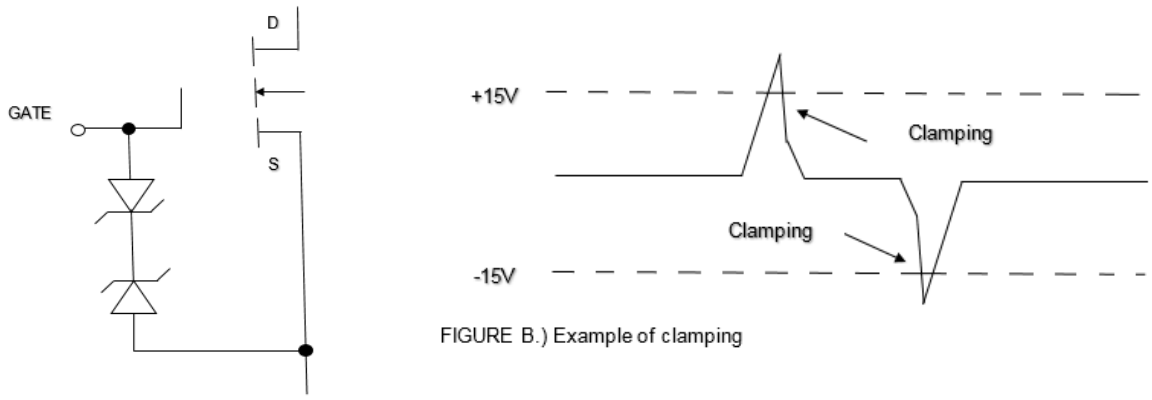


FIGURE B.) Example of clamping

Self-check exercise

1. Name THREE methods for the safe handling of CMOS integrated circuits
2. What is the output of the 7812 voltage regulator?
3. Show by means of a neat, labelled sketch how the terminals of a 7812 voltage regulator should be connected
4. Mention THREE ways in which ESD damage can occur.
5. Explain what you understand by the term electrostatic discharge.
6. What is the basic difference in construction between the base of a bipolar transistor and the gate of a CMOS field-effect transistor?
7. Indicate if the following is **TRUE** or **FALSE**
   2.1.) Some integrated circuits use diodes connected in a forward-biased configuration between the input (gate) and the positive supply to protect the circuits.
   2.2.) Power dissipation in the CMOS circuits is exceeded by the crossover conditions at switching.
   2.3.) MOSFET transistors use less power than the bipolar transistors but are a little slower as far as switching rate is concerned.
8. In which bias direction are the diodes connected in a Zener-diode gate protection?
9. List the procedures to consider when handling C-MOS integrated circuits.

4.2 TRANSDUCERS

4.2.1 Introduction

A transducer ***is a device that converts one form of energy to another form of energy***. In an ***electronically controlled device***, the transducer must Convert another form of energy into an electrical signal proportional to the input signal. The input forms of energy can be Optical, thermal, chemical mechanical, or even electrical energy.

Most transducers can be classified into two groups :	
I.)	Passive transducers
II.)	Self-generating transducers.

Table 4.1

Characteristics of Passive transducers	Characteristics of Self-generating transducers
Passive transducers need electrical power to operate (resistive, inductive, and capacitive transducers). You must, however, bear in mind that a resistor does not require electrical power to determine its resistance, but a resistive transducer needs electrical power to convert the input form of energy to electrical current or voltage	Self-generating transducers generate an emf as soon as they are stimulated by some form of energy and therefore require no external power. A photo-voltaic cell voyage is generated as soon as the cell is stimulated by light energy.

Table 4.2

Transducers can be further classified according to their operating principle :

- Resistive
- Inductive
- Capacitive
- Voltage or Current
- Photo–electric- transducers



QUESTIONS TO CONSIDER WHEN SELECTING A TRANSDUCER

- What is the ***physical quantity*** that should be measured?
- Which ***converting principle*** is most suitable (resistive, inductive)
- Which ***extent of accuracy or reliability*** is Required?

To ***answer the first question***, one should determine which type and range of measurements should be taken. The ***answer to the second question*** requires that

the input and output characteristics of the transducer must be adaptable to the rest of the -system. These two questions can be answered by choosing accuracy tolerance in most cases.

Factors Influencing accuracy of s transducer are :
The adaptability with associated equipment.
The fundamental transducer parameters.
Physical conditions. Temperature, pressure, etc.
Environmental conditions, e.g., heat, humidity, etc.
The ratio of the transducer.

Table 4.3

RESISTIVE TRANSDUCERS

Resistive transducers use the change in resistance to convert other forms of energy like displacement, heat, light, power, torque, humidity, etc., to a proportional current or voltage. Some resistive transducers are looked at closely now.

4.2.2 THE STRAIN GAUGE

THE OPERATION

- The ***strain gauge*** is an example of a ***passive transducer*** that converts mechanical displacement to a change in resistance
- The ***wire resistor or metal foil changes length*** when the strain gauge object is placed under stress
- This ***change in resistance is proportional to the applied voltage*** and is measured by a Wheatstone bridge

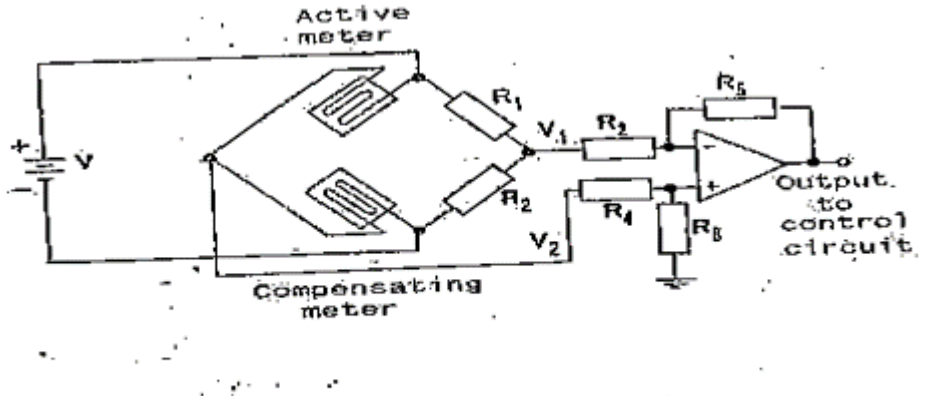


FIGURE 4.1 **Strain gauge in a Wheatstone bridge**

The sensitivity of the strain gauge is called the gauge factor (K) and is defined as the unit change in resistance per unit change in length:

$$\textbf{Gauage Factor } K = \frac{\Delta R/R}{\Delta l/l}$$

$Where\ R - gauge\ resistance$

$\Delta R = Change\ in\ guage\ resistance$

$\Delta l = Change\ in\ length$

$l = normal\ length$

The term $\Delta l/l$ **is the stress** σ **so that :**

$$K = \frac{\Delta R/R}{\sigma}$$

$Where\ \boldsymbol{\sigma} =$
stress in the side wards direction $,$ ***The resistance cahnge*** $\Delta\ R\ of\ a\ conductor\ with\ length\ l\ can\ be\ calculated\ by$

$$\frac{\rho\ x\ l}{\pi d^2/4}$$

$Where\ \rho = specific\ resistance\ of\ conductor$

$l = length\ of\ conductor$

$d = diameter\ of\ conductor$

Hooke's law determines that :

$$\rho = \frac{s}{E}$$

$Where\ \rho = stress\ , \Delta l/l (no\ units)$

$s = strain\ , kg/cm^3$

$E = Youngs\ modulus\ , kg/cm^2$

Example 4.1

A Resistor strain gauge has a stress factor of 2 and is pasted on a steel object under the strain of $1200kg/cm^2$. The modulus of elasticity for steel is approximately $2{,}1\ x\ 10^6$ kg/cm^2.

Calculate the strain gauge percentage change in resistance, ΔR, because of the strain.

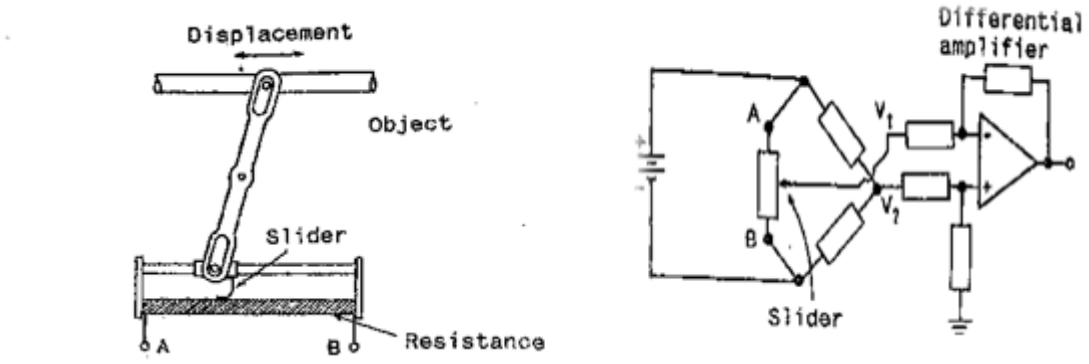
Solution:

$$\sigma = \frac{\Delta l}{l} = \frac{s}{E} = \frac{1200}{1{,}2\ x\ 10^6} = 5{,}714\ x\ 10^{-4}$$

$$\frac{\Delta R}{R} = K\ \sigma = 2\ x\ 5{,}7\ x\ 10^{-4} = 1{,}1\ x\ 10^{-3}\ , therefore\ \%\ change = 1{,}1x10^{-3}x\ 100 = 0{,}11\%$$

4.2.3 THE LINEAR ADJUSTABLE POTENTIOMETER

A linear adjustable potentiometer is used for measuring linear displacement is shown below;

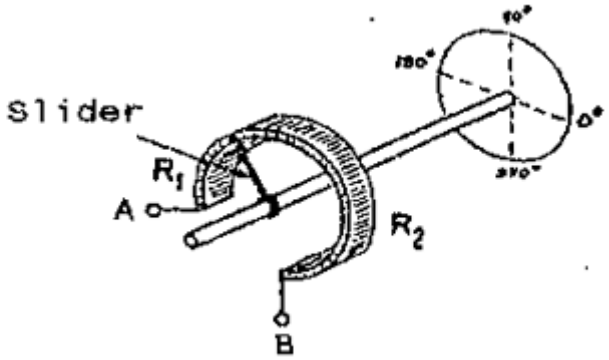


THE OPERATION

- As soon ***as the object moves, the slider is adjusted*** with the aid of lever action. A voltage supply is applied across pins A and B, and the voltage drop, proportional ***to the size of the movement, is measured*** at the slider
- The potentiometer can be connected in a ***Wheatstone bridge, as shown in the above FIGURE.*** The bridge is in balance as soon as the slider is in the center position
- As ***soon as the slider moves to A or B, V1 and V2*** will differ
- The difference is proportional to the amount of displacement; the output polarity indicates the direction of displacement

4.2.4 THE ROTATING POTENTIOMETER

The rotating potentiometer can convert the angular position of an axle or any other object to a proportional voltage, study the following diagram of a rotating potentiometer below;



THE OPERATION

- ***The potentiometer is connected to the Wheatstone bri***dge the same as the linear potentiometer, and the operation is the same. R1 is the resistor between the slider and ***pin A, and R2 is the resistor between the slider and pin*** B
- The single-turn potentiometer can rotate at an angle of approximately 300° and has a low resolution
- ***The Resolution of a potentiometer can be defined as the relation of the voltage*** drop across two adjacent turns to the total ***voltage drops or the relation of one turn to the total number of*** **resistor** ***turns***

$$Resolution = \frac{voltage\ drop\ across\ two\ adjacent\ turns}{total\ voltage\ drop}$$

$$or\ resolution = \frac{1}{number\ of\ turns}$$

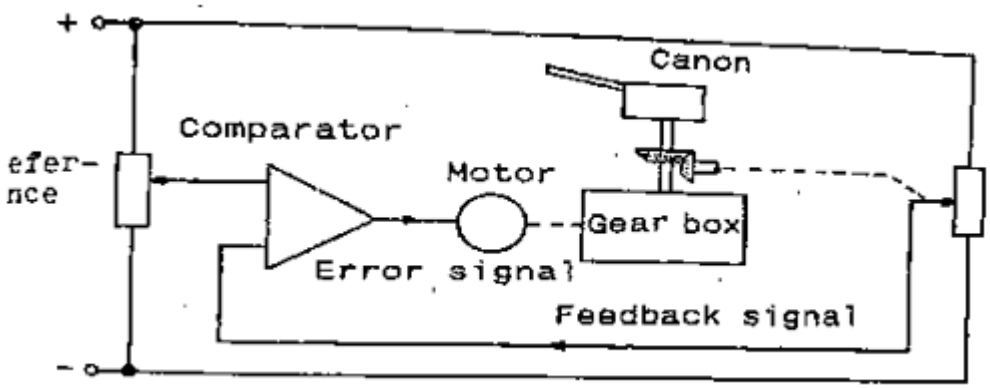
Example 4.2

Calculate the percentage resolution of a potentiometer with 500 turns

Solution: $\%\ Resolution = \frac{1}{500}\ x\ 100 = 0{,}2\ \%$

Thus, ***the total Voltage drop is divided into 500*** voltage drops and the voltage changes by 0,2% between each adjacent turn. ***A potentiometer with more turns or a Potentiometer with more revolutions connected to a gearbox must be used to get a better resolution***. Figure 9.5 Shows a self-balancing potentiometer that is used in a ***Canon turret.***

4.2.5 THE SELF-BALANCING POTENTIOMER

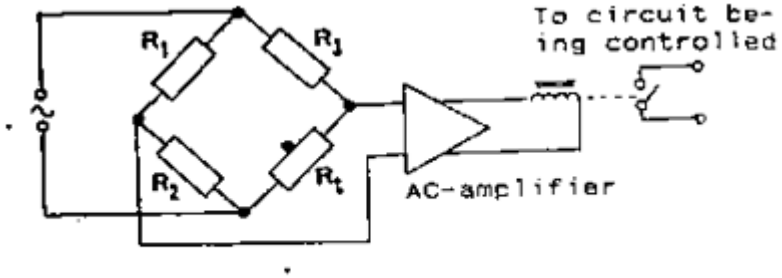


THE OPERATION

- The ***slider of the feedback potentiometer can*** be adjusted mechanically by the axle of the ***canon turret and sends a feedback*** signal to the comparator
- The comparator compares the feedback signal with the reference signal, adjusted manually
- If ***there is a difference, the error signal is given*** and applied to the motor
- The motor rotates the gearbox and Canon, respectively, which rotates the axle of the feedback potentiometer in return
- As the ***Canon nears its required position***, the error signal becomes smaller
- As soon as the canon reaches its required position, the error ***signal is zero, and*** the motor will stop

4.2.6 THE THERMISTOR

The thermistor is known for its function and control of temperature. It has a large resistance change per degree change in temperature. A typical thermistor with a resistance of 2kOhm at 25°C and a temperature- coefficient of 2,9%/°C will maintain a resistance change of 78 Ohms / °C.
A simple temperature control circuit that uses alternating current is shown below



THE OPERATION

- The ***resistor values are calculated so*** that the bridge is in balance at the required temperature
- ***Assume that the thermistor has a value of 10 k Ohm*** at a required temperature of 100°C, then R_3 must = 10 k Ohm and $R_1 = R_3$
- As ***soon as the temperature drops, the resistance of the thermistor*** increases and unbalances the bridge
- The ***thermistor resistance and temperature are inverse proportional,*** and therefore, the thermistor has a negative temperature coefficient (NTC).

The value of the thermistor resistance R_1 is calculated as follows:

$$R_t = Ae^{\frac{B}{T}}$$

$$Where\ A\ and\ B\ = Constants$$
$$e = the\ natural\ logarithm\ (2{,}718)$$
$$\boldsymbol{T = thermo - dynamic\ temperature\ in\ degrees\ Kelvin}$$



Example 4.3

A thermistor has a value of 10kOhm at 10°C and 1kOhm at 30 °.

Calculate:

a.) The values of the constants A and B

b.) The value of the thermistor at 25 °C



Solution:

$R_{t1}: 10\ x\ 10^3 = A^{\frac{B}{283}}$ ---------- (1) ($T = 273 + 10$)

$R_{t2}: 1\ x\ 10^3 = A^{\frac{B}{303}}$ ---------- (1) ($T = 273 + 30$)

$$(1) + (2)\ \frac{10\ x\ 10^3}{1\ x\ 10^3} = \frac{Ae^{\frac{B}{283}}}{Ae^{\frac{B}{303}}}$$

$$10 = e^{\frac{B}{283} - \frac{B}{303}}$$

$$10 = e^{B\left(\frac{1}{283} - \frac{1}{303}\right)}$$

$$\log 10 = B\ (2{,}33\ x\ 10^{-4})(\log e)$$

$$B = 9872{,}2$$

$$Replace\ B\ in\ (20 : 1\ x\ 10^{3} = \ Ae^{9872{,}2/303}$$

$$A = 7{,}079\ x\ 10^{-12}$$

$$The\ Value\ of\ R_1\ at\ 25\ °C :$$

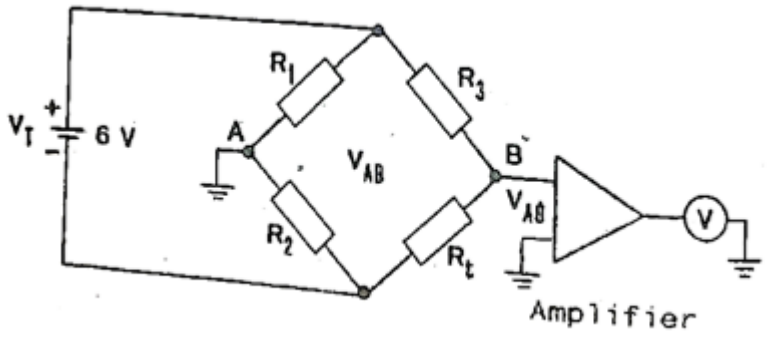
$$R_t = \ A_{e^{B/T}} = 7{,}079\ x\ 10^{-12}\ e^{9872{,}2/298} \qquad (T = 273 + 25\ )$$

$$= 1{,}727\ k\ Ohms$$

Example 4.4

In the figure below, a thermistor bridge with a DC–bias is connected to an amplifier and a voltmeter. The circuit must be in balance at 25 °C. If the value of the thermistor is 10kOhm at 25 °C and constant B = 3600, Calculate:

a.) The values for R_1, R_2, R_3.
b.) The gain of the amplifier at 30 °C to produces a reading of 10V on the output.



Solution:

a.) To balance the bridge at 25 degrees Celsius, the following must be : $R_3 = \ R_t = 10k\ Ohm\ and\ R_1 = \ R_2$. To reach uniformity R_1 and R_2 can also be equal to 10kOhm.

$$Therefore\ R_1 = \ R_2 = \ R_3 = 10kOhm$$

b.) Firstly, determine the value of constant A.

$$R_t = \ Ae^{B/T}$$

$$10\ x\ 10^{3} = \ Ae^{3600/298}$$

$A = 0{,}0567$

At 30 Degrees Celsius the value of the thermistor is

$$R_t = 0{,}0567\ e^{3600/303} = 8{,}192\text{kOhm}$$

To calculate the gain of the amplified, calculate the input voltage.

$$V_{AB} = V_A - V_B$$

$$V_A = \frac{R_2}{R_1 + R_2}(V_T)$$

$$V_A = \frac{10kOhm}{10kOhm + 10kOhm}(6V) = \mathbf{3\ V}$$

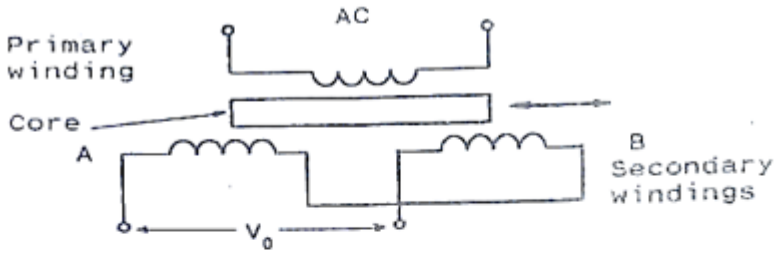
$$V_B = \frac{R_t}{R_t + R_3}(V_T) = \frac{8{,}192kOhm}{8{,}192kOhm + 10\ kOhm}(6V) = \mathbf{2{,}7\ V}$$

Therefore $V_{AB} = V_A - V_B = 3 - 2{,}7 = \mathbf{0{,}3V}$

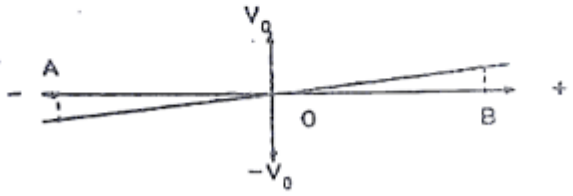
$$Gain\ of\ amplifier\ A_v = \frac{V_O}{V_i} = \frac{10V}{0{,}3V} = 33$$

4.2.7 THE LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT)

The linear variable differential transformer can measure the power displacement of the transformer's ferromagnetic core.



1.) The LVDT



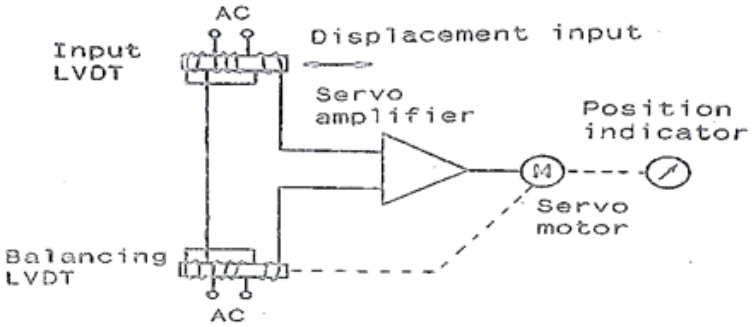
2.) Phase diagram

THE OPERATION

- The ***transformer consists of a primary winding*** and two secondary windings with an equal amount of turns, but which are ***connected in such a way that their voltages*** oppose each other
- With ***the core in the center, equal voltages will be indorsed*** in the secondary windings because ***they oppose each other***, the output V_o will be 0V
- As soon as an external voltage moves the core to the left (A), more 'magnetic flux ***develops in the left winding than in the right*** winding (B)
- Therefore, the **induced voltage in winding A will be greater** than that in winding B
- Therefore, the size ***of the output voltage is equal to the difference*** between the two secondary voltages, and it is in phase with the voltage of winding A. V_o is ***proportional to the amount of displacement*** and the phase of V_o is an indication of the direction in which it moves



The output of the differential transformer can serve as a closed-loop servosystem. FIGURE below shows how two LVDTs can be used.



THE OPERATION

- The servo-amplifier output is zero with both LVDTs in the reference position
- When **the LVDT core moves away from** the reference position, an output voltage is produced to the servo-amplifier
- The ***core of the balancing LVDT*** moves until the output Signal is zero again
- This way, the indicator can indicate ***the amount of displacement that occurred***

4.2.8 THE TACHO – GENERATOR

Tacho- generators are also known as tachometers or tachometer generators and are mostly used in closed-loop motor speed control systems. The anchor of the tacho- generator is connected to the motor's axle mechanically and invokes a voltage V proportional to the motor's speed. This voltage becomes feedback and compares with the reference voltage V_r. The difference between the two voltages is the error signal used to adapt the motor's speed again.[1]

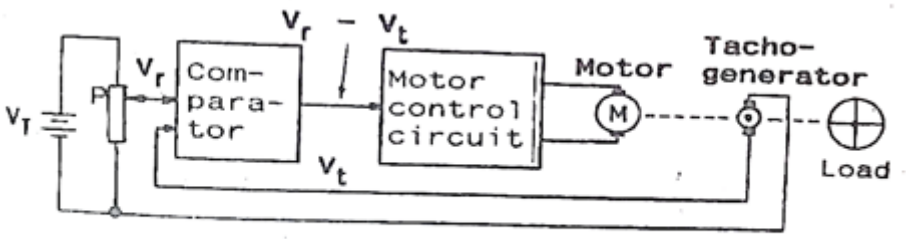
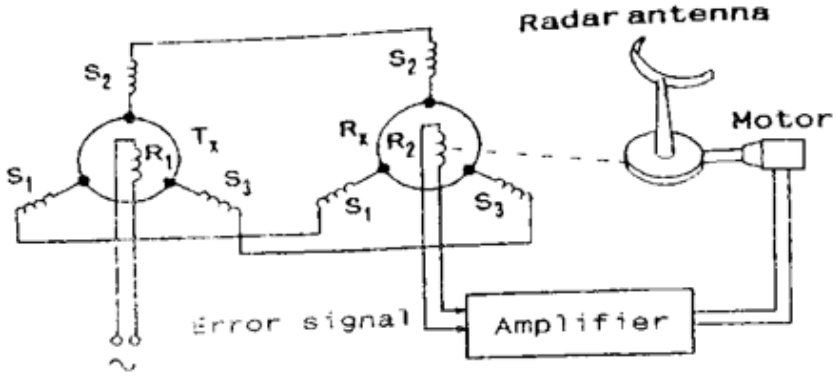


FIGURE 9.10 The Tacho- generator in a closed loop system

4.2.9 THE SYNCHRONOUS MOTOR

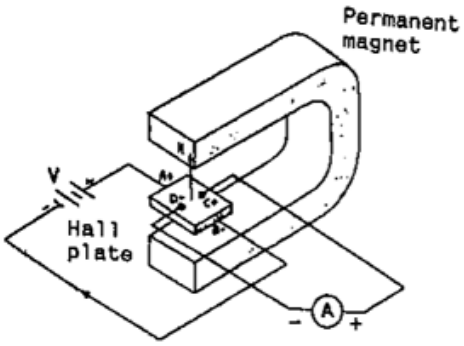
Synchronous ***motors rotate inductive transducers*** and are usually used in pairs to synchronise a transmitter and a receiver. They ***look like small electrical motors*** and c***onsist of three stator windings*** w***rapped electrical 120°, a rotor winding*** connected to two collector rings. ***FIGURE below*** shows ***a synchronous motor pair*** that controls the position of a radar antenna. It is once again a closed-loop system and uses an AC power supply.



VOLTAGE AND CURRENT TRANSDUCERS	INDUCTIVE TRANSDUCERS
Voltage and current transducers invoke an electric voltage or current as soon as they are stimulated by physical energy. Some transducers are passive transducers like the Hall-effect transducer and photo-emitting cell. In contrast, the photo voltaic cell and thermo-couple are classified as self-generating transducers.	Inductive transducers use inductive reactance to convert other forms of energy to a proportional electric signal.

4.2.10 THE HALL-EFFECT TRANSDUCER

The Hall-effect transducer ***is a recording head used to observe magnetic flux and current.*** A potential difference ***is developed across a germanium semi-conductor*** as soon as magnetic flux reacts upon an applied current.

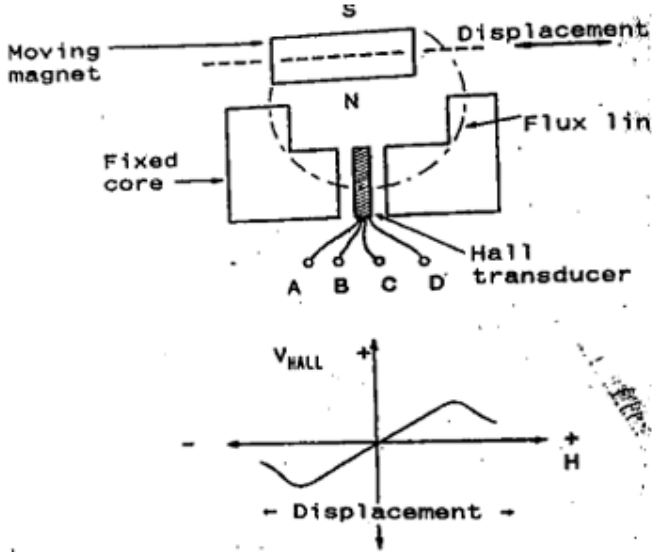


- The Hall-effect transducer consists of a rectangular Germanium semi-conductor plate with two pairs of electrodes, A+, B-, C+, and D-
- A battery sends a direct current between A and B while the magnet invokes a Magnetic field H in the light gap
- The current I and field H induce DC-voltage (V_{HALL}) between C and D so that

$$V_{HALL=}\ CIH$$

Where C = a constant which is dependent on the material used.

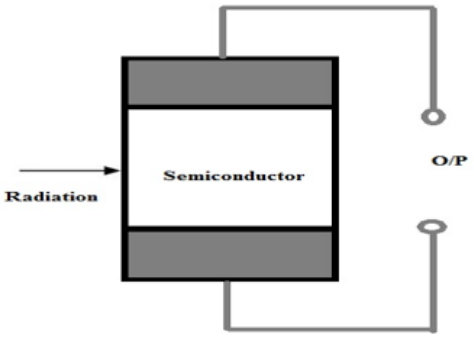
V_{HALL} is ***proportional to the magnetic field H and makes*** this method suitable for measuring magnetic fields. It can ***also measure the power (Watts)*** in DC- and AC circuits. A ***very Sensitive Hall–effect displacement*** transducer is shown in figure below



If ***the magnet is in the center position***, there is no flux and $V_{HALL} = 0$. As soon as the magnet ***moves away from the center position, the*** size and polarity of V_{HALL} will give an indication of the size and direction of the displacement.

4.2.11 PHOTO-ELECTRIC TRANSDUCERS

Photo- electric transducer ***converts light energy to electric energy***. Some of these transducers operate on the resistive principle like the Light-dependent transistor. In contrast, others operate on voltage and current principles like the photo-voltaic cell, photo diode, photo transistor, and light-activated SCR. Infrared detectors convert body temperature to electric signals, and the optic fibre is a conductor of light.

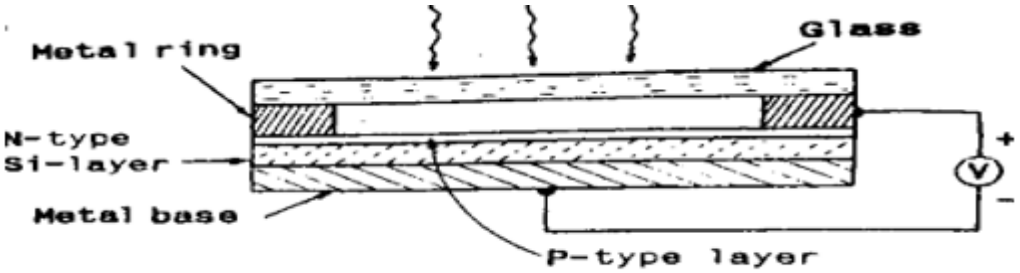


4.2.12 THE PHOTO-VOLTAIC CELL

The photo-voltaic cell is a solar silicon cell that converts the sun's energy to electric power. The cell consists of a very thin slide p-type silicon crystal with a thin layer n-type diffusion layer. The p-type silicon slide is so thin that light can shine through it and reaches the PN junction. An emf develops across this junction as soon as the light shines. Several cells can be connected in series or parallel to produce the required output voltage and current.

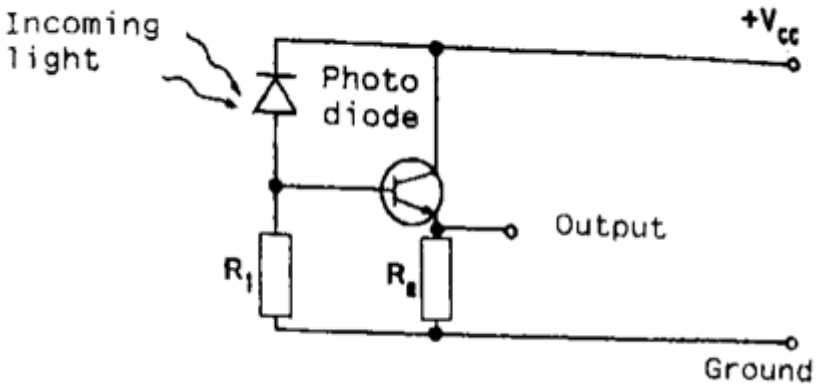
Photo-voltaic cells are used in:

- Battery-less calculators
- Solar heat devices
- Satellite generators



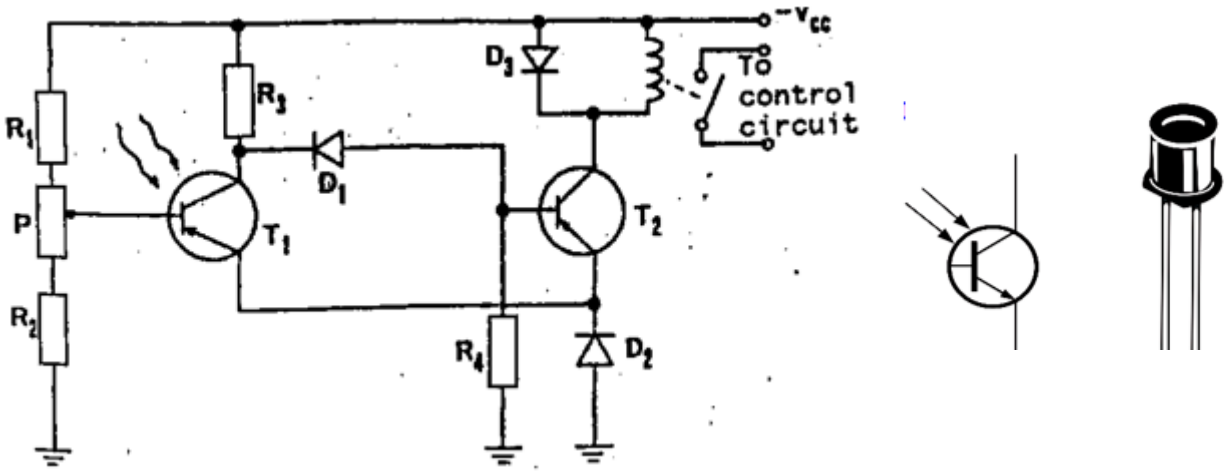
4.2.13 THE PHOTO DIODE

- The photo diode is a semi-conductor PN-junction component of which the surface is in the reverse bias region
- Using this region is advantageous because the inverse current increases linear with the incoming light
- This current, however, is too small for direct application and, therefore, must first be amplified
- However, an advantage of the component is its compactness. Photo diodes are used in infra-red detectors devices that use optic fibre. FIGURE below Shows the photo diode in a common collector amplifier



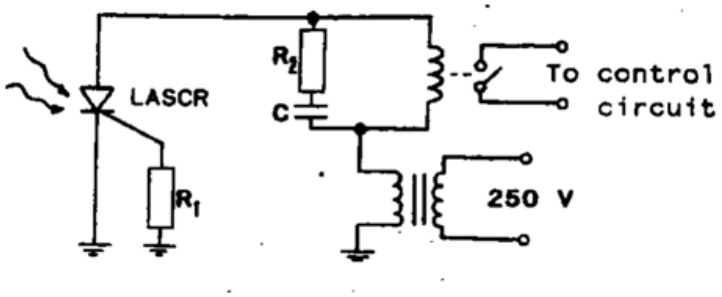
4.2.14 THE PHOTO TRANSISTOR

FIGURE 9.18 ***shows a Schmitt-trigger circuit which includes*** a photo transistor. The relay is in series ***with a transistor T_2 and will be activated as soon as T_2 switches on*** T_2 will switch on as soon as the light intensity drops. Potentiometer ***P*** adjusts the bias of T_1 to compensate for temperature and surrounding light.



4.2.15 THE LIGHT-ACTIVATED SILICON CONTROLLED RECTIFIER (LASCR)

The light-activated SCR uses a circuit that converts alternating current to direct current. A standard DC relay is used. A capacitor is connected parallel with the relay coil to eliminate the half-wave vibrations. **FIGURE below** shows a relay circuit that the LASCR controls:

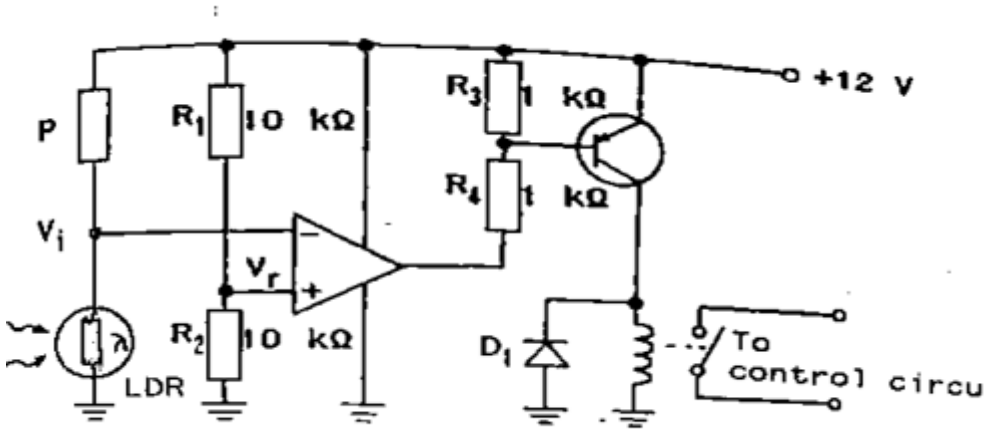


THE OPERATION

- LASCR ***can also be used in an AND-function*** (two LASCRs in series) or an OR-function (two LASCRs in parallel)
- Both components must receive light to pass current to a load in the first case
- In the ***second case, only one or other LASCR needs to receive light*** to pass current to a load

4.2.16 THE LIGHT DEPENDANT RESISTOR

The ***light-dependent resistor (LDR), or photo-conductive cell resistance, reduces with an increase in light intensity*** (typically from 100k Ohms to 100 Ohms). The essential elements of this transducer are the photoconductor which consists ***of Cadmium sulphide*** and ***metal electrodes.***



THE OPERATION

- The LDR is used especially in precision light switches to control, amongst others, streetlights. ***The circuit above switches the relay*** on as soon as the light drops below a certain intensity
- R_1 and R_2 forms ***a voltage divider that produces a reference*** voltage to the operational comparator
- P is ***a precision potentiometer and adjusts*** the input V for different light Intensities
- As soon as the light intensity drops, the resistance of the LDR increases, and V increases
- ***As soon as*** $V_i < V_r$***The comparator*** output goes low and switches on T_1 because of the base bias that increases across R_3
- Current flows through the relay and short circuits the Contacts.
- The circuit will react reversely by only changing the LDR and Potentiometer P. The relay will switch ***on as soon as light rises above a certain intensity***

4.2.17 THE INFRARED DETECTOR

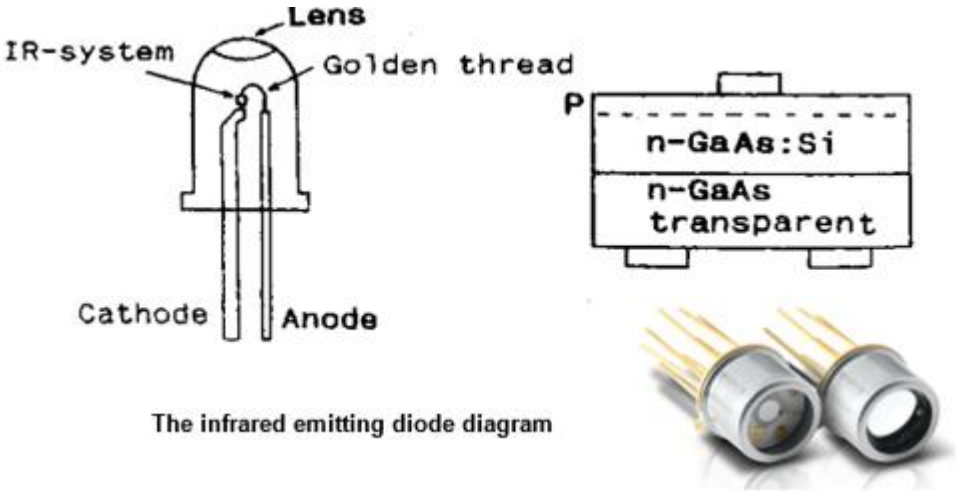
An ***infra-red beam is an invisible beam*** that is invoked by an infra-red emitting diode with a built-in lens to ***concentrate the energy in a narrow beam***. The ***beam is observed by a photo diode with an infra-red filter*** in front of it to keep off normal light. The signal is amplified and can be used to control a relay.

This principle can be applied in the following :

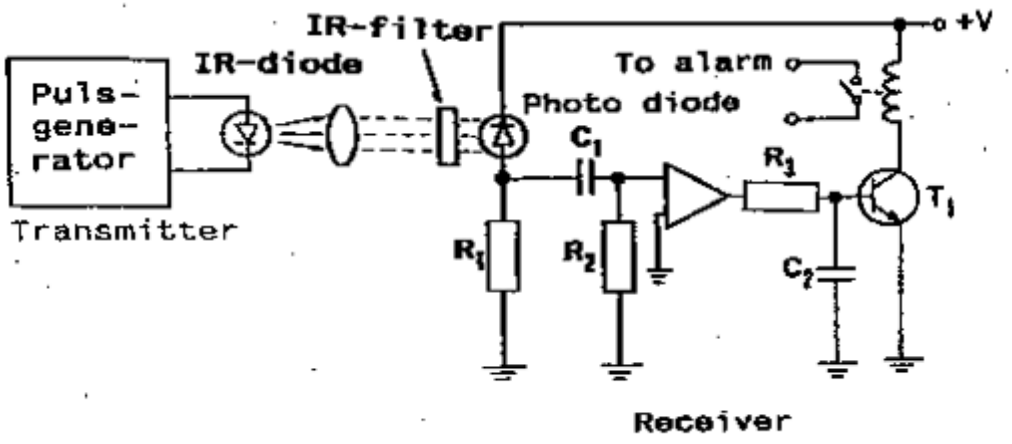
▪ Alarm systems
▪ Automatic door ▪ Openers
▪ Doorbells
▪ Light switches
▪ Distance control circuits
▪ The detection of human ▪ movement

Table 4.4

The infrared beam can be directed directly on the photo diode or reflected on the photo diode using a flat surface like a wall.



The infrared emitting diode diagram

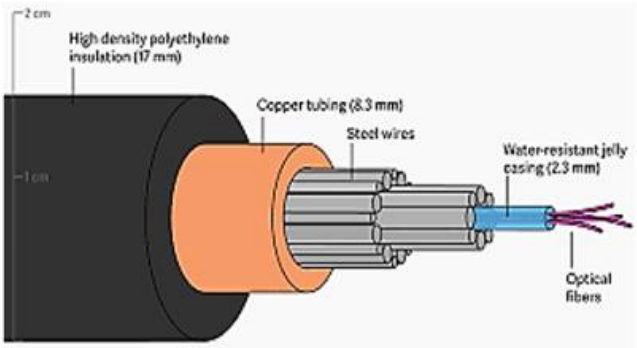


THE OPERATION

- The capacitor C_1 passes the pulses but blocks the slow Variations of surrounding light
- The photo diode conveys the pulses to the amplifier. C_2 Forces the pulses to a smooth direct current which keeps T_1 on and let current flow through the relay
- As soon as the emission of an object or person is interrupted, no signal is conveyed to the transistor and therefore switches it off
- The relay switches to its normal condition so that the contacts short circuits and activates the alarm

4.2.18 THE OPTIC FIBRE

The optic fibre is a conductor of light used in communication systems and medical cameras. As soon as the light is shone at the one side of the optic fibre, it is reproduced using refraction or reflection of the sheath, which reacts as thousands of little mirrors. The fibre is flexible and enables light to be conducted at an angle. The sheath is of glass or plastic and has a high refraction index. If a lot of fibre is composed in a bunch, an object placed on one end can be seen directly on the other end. That is why it is suitable for a medical camera with which a specialist can examine a human stomach.







A transmitter and a receiver are connected to both sides of the optic fibre in communication systems*. The transmitter sends light pulses according to a specific code while the receiver observes and invokes these pulses with a photo-electric transducer. The ends of the optic fibre must be smooth and hooked. This medium reaches the ideal medium because no noise or influences from outside can affect the light signals.*

APPLICATIONS

- **Telecommunications** for transmitting data over long distances in telecom networks
- **Medical imaging** for conducting light in medical devices such as endoscopes and laser surgery equipment.
- **Sensors** for temperature monitoring, pressure sensing and chemical sensing in industry.
- **Military and aero scope** for communication, sensing and navigation system due to their light weight and immunity to electromagnet interference.

SELF CHECK 9

1.) Transducers can be classified into two groups. Outline the groups.
2.) Define the term transducer.
3.) Mention 5 factors that can influence converting accuracy.
4.) Mention 4 converting principles in which transducers are obtainable.
5.) Draw a circuit diagram to show how strain gauges must be connected in a Wheatstone bridge and the differential amplifier that observes the displacement equivalent signal.
6.) Define the term resolution of a wire-wound potentiometer.
7.) Calculate the percentage resolution of a potentiometer with 1000 turns.
8.) Determine for a rotating potentiometer given the following parameters
    8.1.1 Percentage resolution for 0.5V between turns and 24V across the potentiometer.
    8.1.2 Voltage drop between windings for resolution of 0.75% and voltage drop of 20V.
9.) Draw the circuit diagram of a closed-loop system that uses a potentiometer to direct, e.g., a rotating canon.
10.) A distortion gauge with a gauge factor of 3 is pasted to a steel beam under the strain of 1600 kg/cm^3. Calculate the percentage change in resistance, ΔR of the distortion gauge because of the strain.
11.) Calculate the resolutions of a wire-wound potentiometer with 350 turns.
10.) Draw a neat, labelled circuit diagram of a photoelectric transducer that can be used for heat devices or satellite generators.

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